

B4 — Nuclear and Particle Physics Model Solutions, Trinity 2024

Question 1 — Nuclear form factors and electron scattering

Problem. Define $F(q)$ for elastic charged-particle–nucleus scattering; compute it for a uniform sphere; analyse limits and softening of minima; state the relativistic correction; estimate $E_{\min}$ for $R = 3 \text{ fm}$.

(a) The form factor and its evaluation for a uniform sphere

Born amplitude for a point nucleus is the Rutherford amplitude $f_R(q)$. For an extended charge density $\rho(\mathbf{r})$ with $\int \rho \, d^3r = Ze$:

$$f(\mathbf{q}) = f_R(q) F(\mathbf{q}), \quad F(\mathbf{q}) = \frac{1}{Ze} \int \rho(\mathbf{r}) e^{i\mathbf{q}\cdot\mathbf{r}/\hbar} d^3r. \quad (1)$$

Differential cross-section: $d\sigma/d\Omega = (d\sigma/d\Omega)_{\text{Ruth}} |F(q)|^2$, with $F(0) = 1$.

Spherical symmetry, polar axis along $\mathbf{q}$, angular integral $\int_{-1}^1 e^{iqr \cos \theta/\hbar} d \cos \theta = 2 \sin(qr/\hbar)/(qr/\hbar)$:

$$F(q) = \frac{4\pi}{Ze} \int_0^\infty \rho(r) \frac{\sin(qr/\hbar)}{qr/\hbar} r^2 dr. \quad (2)$$

Set $\hbar = 1$ henceforth.

Uniform sphere: $\rho = \rho_0 \Theta(R - r)$, $Ze = (4/3)\pi R^3 \rho_0$. Substitute into (2):

$$F(q) = \frac{4\pi\rho_0}{Ze} \int_0^R \frac{\sin(qr)}{qr} r^2 dr.$$

Change variable $u = qr$:

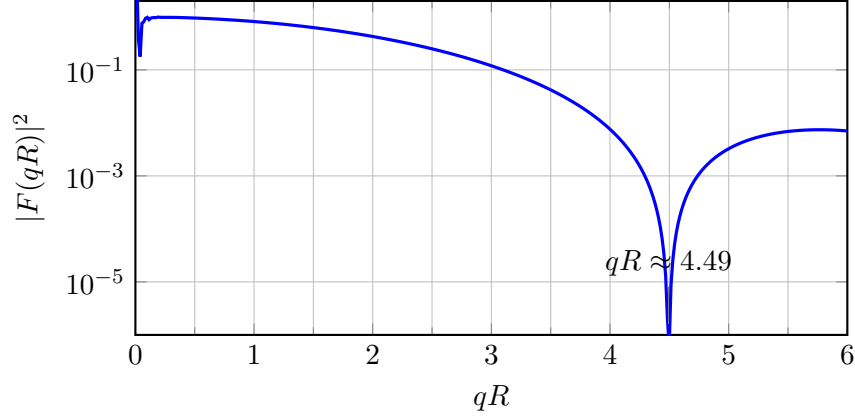
$$F(q) = \frac{3}{R^3 q^3} \int_0^{qR} u \sin u \, du.$$

Integrate by parts: $\int_0^x u \sin u \, du = \sin x - x \cos x$:

$$\boxed{F(q) = \frac{3}{(qR)^3} [\sin(qR) - qR \cos(qR)] = \frac{3 j_1(qR)}{qR}.} \quad (3)$$

Zeros: $\tan(qR) = qR$, i.e. $qR \approx 4.493, 7.725, \dots$; one minimum within $0 < qR < 6$.

Sketch of $|F(q)|^2$.



(b) Limits $q \rightarrow 0$, $q \rightarrow \infty$, and softening of minima

Small- qR expansion: $\sin(qR) - qR \cos(qR) = (qR)^3/3 - (qR)^5/30 + \dots$:

$$F(q) = 1 - \frac{1}{10}(qR)^2 + \dots = 1 - \frac{1}{6}q^2 \langle r^2 \rangle + \dots$$

Read off $\langle r^2 \rangle = \frac{3}{5}R^2$. Hence

$$\boxed{\lim_{q \rightarrow 0} F(q) = 1.} \quad (4)$$

Charge conservation: a long-wavelength probe sees only Ze .

Large- qR : the bracket in (3) is bounded while $(qR)^3 \rightarrow \infty$:

$$\boxed{\lim_{q \rightarrow \infty} F(q) = 0.} \quad (5)$$

Short-wavelength contributions interfere destructively over the volume.

Softening of the minima. The exact zero is an artefact of the hard edge. Filling-in mechanisms:

(i) Woods–Saxon surface diffuseness $a \sim 0.5$ fm; (ii) intrinsic proton form factor $F_p(q)$ multiplying $F(q)$; (iii) Coulomb distortion / two-photon exchange (beyond first Born); (iv) inelastic excitation and recoil. Surface diffuseness is the dominant effect.

(c) Relativistic correction to Rutherford scattering

Rutherford assumes non-relativistic spinless point charges in a Coulomb field. For $E \gtrsim m_e c^2$ two effects enter.

Apply $E^2 = p^2 c^2 + m_e^2 c^4$:

$$E \neq p^2/2m_e \quad (\text{relativistic kinematics}).$$

Include Dirac spin coupling to the magnetic field in the nuclear rest frame (Mott):

$$\left(\frac{d\sigma}{d\Omega} \right)_{\text{Mott}} = \left(\frac{d\sigma}{d\Omega} \right)_{\text{Ruth}} [1 - \beta^2 \sin^2(\theta/2)]. \quad (6)$$

Backward suppression as $\beta \rightarrow 1$: helicity conservation forbids the 180° flip in the $m_e \rightarrow 0$ limit. (Nuclear recoil and $|F(q)|^2$ further required.)

(d) Determining $\rho(r)$ and the minimum energy to resolve $R = 3 \text{ fm}$

Procedure. Measure $d\sigma/d\Omega(\theta)$ at fixed E_{beam} ; divide by Mott (with recoil) to extract $|F(q)|^2$; fit $\rho(r; \{\alpha_i\})$ (Fermi / sum-of-Gaussians) so that (2) reproduces the data; inverse Fourier transform yields $\rho(r)$.

To resolve the first diffraction minimum:

$$qR \gtrsim 4.49.$$

Solve for q at $R = 3 \text{ fm}$ using $\hbar c = 197 \text{ MeV fm}$:

$$q \gtrsim \frac{4.49 \times 197 \text{ MeV fm}}{3 \text{ fm}} \approx 295 \text{ MeV/c}.$$

Ultra-relativistic backscattering: $q_{\text{max}} \simeq 2p \approx 2E/c$:

$$E_{\text{min}} \simeq \frac{1}{2} q_{\text{max}} c = \frac{1}{2} \times 295 \text{ MeV}.$$

$$\boxed{E_{\text{min}} \approx 150 \text{ MeV} .} \quad (7)$$

Weaker criterion $qR \sim \pi$ gives $E_{\text{min}} \approx 100 \text{ MeV}$.

Question 2 — Semi-Empirical Mass Formula and the nuclear force

Problem. Origin of the A and $A^{2/3}$ terms; uncertainty estimate of p^2 ; size of a $100\times$ -Coulomb two-nucleon bound state. Exchange-force mass; pn peaks. Stable Z on the odd- A parabola; pairing and even- A stability.

(a) Saturation of the nuclear force, the volume and surface terms

Empirical: $r = r_0 A^{1/3}$ with $r_0 \approx 1.2 \text{ fm}$, so $V \propto A$ and central density is constant (saturation); $BE/A \sim 8 \text{ MeV}$. Both demand short-range NN force with hard core. Long-range force would give $BE \propto A(A-1)/2 \propto A^2$, contradicting saturation.

Volume term $a_V A$: each interior nucleon has the same number of nearest neighbours. Surface term $-a_S A^{2/3}$: surface nucleons lack neighbours, count $\propto 4\pi R^2 \propto A^{2/3}$, sign negative.

Uncertainty-principle estimate of p^2 . Confinement $\Delta x \sim r$:

$$p^2 \sim (\Delta p)^2 \sim \frac{\hbar^2}{r^2}. \quad (8)$$

Kinetic energy $T = p^2/2m_N \sim \hbar^2/(2m_N r^2)$ rises as $r \rightarrow 0$, opposing attraction.

Two-nucleon bound state with $V = -100 e^2/(4\pi\epsilon_0 r)$. Effective coupling $\alpha_{\text{eff}} = 100\alpha = 100/137 \approx 0.73$; reduced mass $\mu = m_N/2$.

Variational energy:

$$E(r) = \frac{\hbar^2}{2\mu r^2} - \frac{100\alpha \hbar c}{r}.$$

Differentiate and set $dE/dr = 0$:

$$-\frac{\hbar^2}{\mu r^3} + \frac{100\alpha \hbar c}{r^2} = 0.$$

Solve for r :

$$r_{\min} = \frac{\hbar^2}{\mu (100\alpha) \hbar c} = \frac{\hbar c}{(\mu c^2) (100\alpha)}.$$

Substitute $\mu c^2 = m_N c^2 / 2 \approx 469.5 \text{ MeV}$, $\hbar c = 197 \text{ MeV fm}$, $100\alpha \approx 0.73$:

$$r_{\min} = \frac{197}{469.5 \times 0.73} \text{ fm} = \frac{197}{342.7} \text{ fm}.$$

$$\boxed{r_{\min} \approx 0.6 \text{ fm} .} \quad (9)$$

Order of the deuteron radius ($\sim 2 \text{ fm}$): a $100\times$ -Coulomb model captures the strong-force scale to a factor of a few.

(b) Exchange forces, Yukawa range, and the pn scattering peaks

Virtual exchange of mass m_X , lifetime $\Delta t \sim \hbar / (m_X c^2)$, range $r \lesssim c \Delta t$:

$$R_{\text{Yuk}} = \frac{\hbar}{m_X c}. \quad (10)$$

Identify with $r_0 = 1.2 \text{ fm}$:

$$m_X c^2 \sim \frac{\hbar c}{r_0} = \frac{197 \text{ MeV fm}}{1.2 \text{ fm}}.$$

$$\boxed{m_X c^2 \approx 160 \text{ MeV} .} \quad (11)$$

Compare $m_\pi \approx 140 \text{ MeV}/c^2$ (Yukawa).

Forward and backward peaks in pn scattering. Charge-exchange $\pi^\pm$ swaps proton and neutron identities:

$$p + n \rightarrow n + p \quad (\pi^\pm \text{ exchange}).$$

A forward neutron in the lab corresponds to a “backward proton” in the original-proton frame, producing a peak at $\theta_p = 180^\circ$. The backward peak is direct evidence of the exchange-force component of the NN interaction.

(c) Mass parabola, β -decay channels, and the role of pairing

For fixed A , BE is quadratic in Z :

$$BE(Z, A) = (Z\text{-indep.}) - a_C \frac{Z^2}{A^{1/3}} - a_A \frac{(Z - A/2)^2}{A} \pm \delta(A, Z). \quad (12)$$

Atomic mass $M(Z, A) = Zm_p + (A - Z)m_n - BE/c^2$ is also quadratic in Z . For odd- Z even- N ($o-e$) isobars $\delta = 0$, so all such isobars lie on a single parabola.

Stationary point:

$$\left. \frac{\partial BE}{\partial Z} \right|_A = -\frac{2a_C Z}{A^{1/3}} + a_A \frac{A - 2Z}{A} - (m_n - m_p)c^2 = 0.$$

Solve for Z :

$$\boxed{Z_{\min}(A) \approx \frac{A/2}{1 + (a_C/a_A) A^{2/3}} .} \quad (13)$$

Using $a_C \approx 0.71 \text{ MeV}$, $a_A \approx 23 \text{ MeV}$: light $A \Rightarrow Z_{\min} \approx A/2$ (i.e. $N = Z$); heavy $A \sim 200 \Rightarrow Z_{\min}/A \approx 0.4$.

Three decay channels for unstable isobars.

1. β^- : $n \rightarrow p + e^- + \bar{\nu}_e$ ($Z < Z_{\min}$, $\Delta Z = +1$).
2. β^+ : $p \rightarrow n + e^+ + \nu_e$ ($Z > Z_{\min}$, $Q > 2m_e c^2 \approx 1.022 \text{ MeV}$, $\Delta Z = -1$).
3. Electron capture: $p + e_{\text{atomic}}^- \rightarrow n + \nu_e$ ($Q > 0$, $\Delta Z = -1$); dominates β^+ in heavy nuclei (no $2m_e c^2$ phase-space penalty, large inner-shell overlap).

Pairing for $e-e$ and $o-o$ isobars. Empirically

$$\delta \approx \begin{cases} +a_P/A^{1/2}, & e-e \text{ (more bound)}, \\ 0, & o-e, \\ -a_P/A^{1/2}, & o-o \text{ (less bound)}, \end{cases} \quad (14)$$

$a_P \approx 12 \text{ MeV}$. For even A , $e-e$ and $o-o$ chains lie on *two* parabolae offset by $\sim 2a_P/A^{1/2}$.

Consequence: an $e-e$ isobar can be stable even off the parabola minimum, because the only single- β neighbour is the less-bound $o-o$ isobar one step away. Example $A = 128$: ^{128}Te and ^{128}Xe ($e-e$) both stable, separated by ^{128}I ($o-o$) above; only 2β -decay $^{128}\text{Te} \rightarrow ^{128}\text{Xe} + 2e^- + 2\bar{\nu}_e$ connects them, $\tau \sim 10^{24} \text{ yr}$. Empirical census: ~ 165 stable $e-e$ nuclei, only 4 stable $o-o$ (^2H , ^6Li , ^{10}B , ^{14}N).

Question 3 — The pseudoscalar meson nonet

Problem. Identify the axes of the $J^P = 0^-$ nonet; derive J^P and $Q = I_3 + S/2$ from quark content; show single-body decay forbidden; draw $\pi^0 \rightarrow \gamma\gamma$; estimate Dalitz suppressions; explain $\text{BR}(\eta \rightarrow \gamma\gamma) < \text{BR}(\pi^0 \rightarrow \gamma\gamma)$. Charged π, K decay via virtual W ; long lifetime; why K^0 cannot mimic π^0 .

(a) Axes of the multiplet

Horizontal: I_3 . Vertical: S . Quark assignments: u ($I_3=+1/2, S=0$), d ($I_3=-1/2, S=0$), s ($I_3=0, S=-1$); antiquarks flip sign. The I_3 - S plane is one face of the $\text{SU}(3)_{\text{flav}}$ weight diagram.

(b) Spin, parity, charge formula and the nonet

Quarks are spin-1/2 fermions; for $L = 0$:

$$J = S_{\text{tot}} \in \{0, 1\}.$$

Intrinsic parity of $q\bar{q}$:

$$P = P_q P_{\bar{q}} (-1)^L = (+1)(-1)(+1) = -1.$$

Hence

$$J_{\text{nonet}}^P = 0^- \quad (S_{\text{tot}} = 0, L = 0). \quad (15)$$

Gell-Mann–Nishijima. Quark charges: $Q_u = +2/3, Q_d = Q_s = -1/3$. With $B_q = 1/3$:

$$Q = I_3 + \frac{1}{2}(S + B).$$

For a meson $B = B_q + B_{\bar{q}} = 0$:

$$\boxed{Q_{\text{meson}} = I_3 + \frac{S}{2}}. \quad (16)$$

Check $K^+(I_3 = +1/2, S = +1)$: $Q = 1$. Check $\pi^+(I_3 = +1, S = 0)$: $Q = 1$. Check $\bar{K}^0(I_3 = +1/2, S = -1)$: $Q = 0$.

Why nine mesons. Three flavours: $3 \otimes \bar{3} = 8 \oplus 1$. Octet ($\pi^{\pm,0}, K^{\pm,0,\bar{0}}, \eta_8$) plus singlet η_1 ; physical η, η' mix.

Why none stable. Each has an open channel. π^0, η, η' : electromagnetic $\rightarrow \gamma\gamma$. Charged π, K and K^0 : weak (virtual W). η' : strong $\eta' \rightarrow \eta\pi\pi$. No absolutely conserved quantum number forbids the lightest open channel.

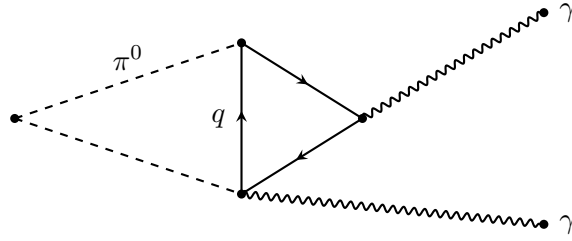
(c) Single-body decay forbidden, $\pi^0 \rightarrow \gamma\gamma$ and Dalitz suppressions

Parent M at rest, single daughter m . Energy: $E_f = Mc^2$. Momentum: $\mathbf{p}_f = 0$, hence $E_f = mc^2$. Equate:

$$M = m.$$

For $M \neq m$ impossible; require ≥ 2 daughters. Lightest pair for π^0 : $\gamma\gamma$.

Feynman diagram for $\pi^0 \rightarrow \gamma\gamma$.



Suppression of Dalitz channels. Each replaced photon $\rightarrow e^+e^-$ adds one γe^+e^- vertex, i.e. one factor of α in the rate.

Single Dalitz:

$$\frac{\Gamma(\pi^0 \rightarrow \gamma e^+ e^-)}{\Gamma(\pi^0 \rightarrow \gamma\gamma)} \sim \alpha \approx \frac{1}{137}. \quad (17)$$

Double Dalitz:

$$\frac{\Gamma(\pi^0 \rightarrow e^+ e^- e^+ e^-)}{\Gamma(\pi^0 \rightarrow \gamma\gamma)} \sim \alpha^2 \approx \frac{1}{1.9 \times 10^4}. \quad (18)$$

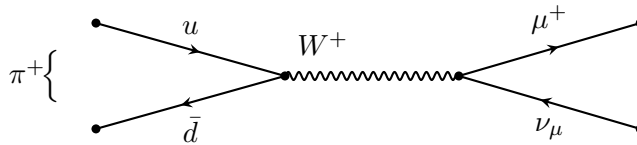
Measured: $\text{BR}(\pi^0 \rightarrow \gamma e^+ e^-) = 1.17\%$, $\text{BR}(\pi^0 \rightarrow e^+ e^- e^+ e^-) = 3.3 \times 10^{-5}$.

Why $\text{BR}(\eta \rightarrow \gamma\gamma) < \text{BR}(\pi^0 \rightarrow \gamma\gamma)$. The partial $\gamma\gamma$ widths are comparable, but the η has additional open channels (e.g. $3\pi^0, \pi^+\pi^-\pi^0, \pi^+\pi^-\gamma$) closed for the lighter π^0 . Hence $\Gamma_\eta \approx 1.3 \text{ keV} \gg \Gamma_{\pi^0} \approx 8 \text{ eV}$:

$$\text{BR}(\eta \rightarrow \gamma\gamma) \approx 39\% < \text{BR}(\pi^0 \rightarrow \gamma\gamma) \approx 99\%. \quad (19)$$

(d) Charged pion and kaon decays; the K^0 case

$\pi^+ = u\bar{d}$ annihilates into virtual $W^+ \rightarrow \ell^+ \nu_\ell$:



Why a heavy W mediates a light decay. The W is virtual; energy-time uncertainty allows $\Delta t \sim \hbar/m_W c^2$; propagator $\sim 1/(q^2 - m_W^2 c^2) \approx -1/(m_W c)^2$ at $q^2 \ll m_W^2 c^2$.

Why the lifetime is long. Two factors.

Suppression by Fermi constant:

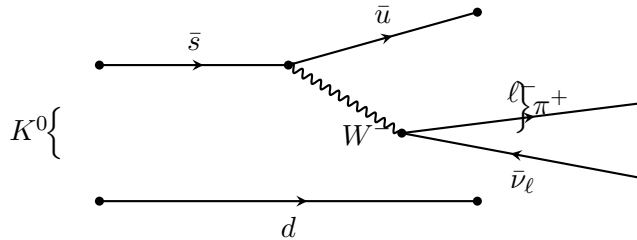
$$\frac{G_F}{\sqrt{2}} = \frac{g_w^2}{8m_W^2 c^4} \quad (\text{small in natural units}).$$

Helicity suppression in $\pi^+ \rightarrow \ell^+ \nu_\ell$ (spin-0 parent, $V-A$ couplings forces same-helicity $\ell^+ \nu_\ell$ that is forbidden by back-to-back angular momentum unless $m_\ell \neq 0$):

$$\Gamma \propto m_\ell^2 (m_\pi^2 - m_\ell^2)^2.$$

Together $\tau_{\pi^\pm} \approx 2.6 \times 10^{-8}$ s; eight orders of magnitude longer than strong decays and than $\tau_{\pi^0} \approx 8 \times 10^{-17}$ s.

Why K^0 cannot use the π^0 mode. $K^0 = d\bar{s}$, $S = +1$. Photon annihilation requires q and $\bar{q}$ of the *same* flavour, which $d\bar{s}$ is not; electromagnetism conserves flavour, so cannot turn $s \leftrightarrow d$. Only the weak interaction can change strangeness ($\Delta S = 1$). Representative semileptonic ($K_{\ell 3}$) diagram for $K^0 \rightarrow \pi^+ \ell^- \bar{\nu}_\ell$:

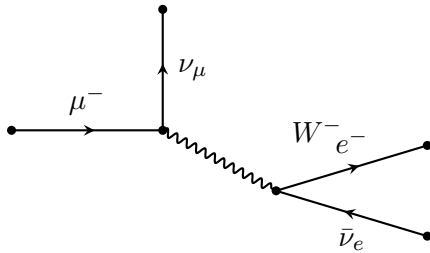


Question 4 — Lepton universality and quark mixing

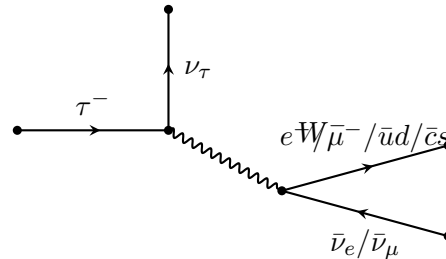
Problem. Diagrams for μ and τ decay; hadronic decays; justify $\Gamma(\tau \rightarrow e \nu \bar{\nu})/\Gamma(\mu \rightarrow e \nu \bar{\nu}) \simeq (m_\tau/m_\mu)^5 (g_\tau/g_\mu)^2$. Evaluate g_τ/g_μ . Quark-level diagrams for n, Λ^0 β -decay; restoration of universality via Cabibbo mixing; minimum quark count; θ_C . Predict τ branching ratios; comment on heavy-quark stability.

(a) Leptonic decays and the m^5 scaling

Feynman diagrams.



(a) $\mu^- \rightarrow e^- \bar{\nu}_e \nu_\mu$



(b) $\tau^- \rightarrow \nu_\tau + W^{*-}$

Hadronic decays. $m_\mu c^2 \approx 106$ MeV $<$ $m_\pi \approx 140$ MeV: muon hadronic decay forbidden. $m_\tau c^2 \approx 1777$ MeV $\gg m_\pi$: $\tau \rightarrow \nu_\tau + \bar{u}d/\bar{c}s$ allowed; quarks hadronise.

The m^5 rule. For $m_\ell \ll m_W$, the W propagator contracts to a four-fermion contact:

$$\mathcal{L}_{\text{eff}} \propto \frac{g_\ell g_e}{m_W^2} (\bar{\nu}_\ell \ell)(\bar{e} \nu_e).$$

Three-body massless phase space scales as m_ℓ^5 (only available scale, G_F has dimension $[\text{E}]^{-2}$). Standard result:

$$\Gamma(\ell^- \rightarrow \nu_\ell e^- \bar{\nu}_e) = \frac{g_\ell^2 g_e^2}{192\pi^3} \frac{m_\ell^5 c^4}{m_W^4 \hbar}. \quad (20)$$

Form the ratio for τ over μ (g_e cancels):

$$\boxed{\frac{\Gamma(\tau^- \rightarrow e^- \bar{\nu}_e \nu_\tau)}{\Gamma(\mu^- \rightarrow e^- \bar{\nu}_e \nu_\mu)} = \left(\frac{m_\tau}{m_\mu}\right)^5 \left(\frac{g_\tau}{g_\mu}\right)^2}. \quad (21)$$

(b) Numerical evaluation of g_τ/g_μ

Widths from lifetimes:

$$\Gamma(\tau \rightarrow e \nu \bar{\nu}) = \frac{\hbar \text{BR}(\tau \rightarrow e \nu \bar{\nu})}{\tau_\tau}, \quad \Gamma(\mu) = \frac{\hbar}{\tau_\mu} (\text{BR} = 1).$$

Form the ratio:

$$\frac{\Gamma_\tau^{(e)}}{\Gamma_\mu} = \frac{\tau_\mu}{\tau_\tau} \text{BR}(\tau \rightarrow e \nu \bar{\nu}).$$

Substitute $\tau_\mu = 2.197 \times 10^{-6} \text{ s}$, $\tau_\tau = 2.91 \times 10^{-13} \text{ s}$, $\text{BR} = 0.178$:

$$\frac{\Gamma_\tau^{(e)}}{\Gamma_\mu} = \frac{2.197 \times 10^{-6}}{2.91 \times 10^{-13}} \times 0.178.$$

Evaluate:

$$\frac{\Gamma_\tau^{(e)}}{\Gamma_\mu} \approx 7.55 \times 10^6 \times 0.178 \approx 1.34 \times 10^6.$$

Mass-ratio factor:

$$\left(\frac{m_\tau}{m_\mu}\right)^5 = \left(\frac{1777.0}{105.66}\right)^5 = (16.82)^5.$$

Evaluate:

$$\left(\frac{m_\tau}{m_\mu}\right)^5 \approx 1.347 \times 10^6.$$

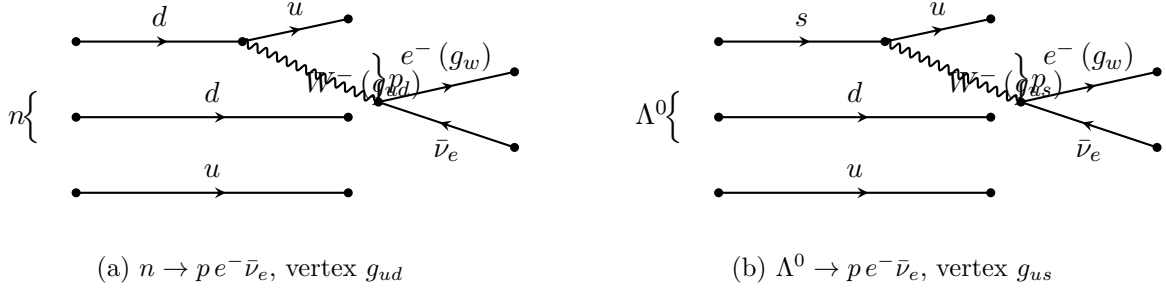
Take the ratio:

$$\left(\frac{g_\tau}{g_\mu}\right)^2 = \frac{1.34 \times 10^6}{1.347 \times 10^6} \approx 0.995.$$

$$\boxed{\frac{g_\tau}{g_\mu} \approx 1.00}. \quad (22)$$

Lepton universality at the W vertex.

(c) Quark-level Feynman diagrams for n and Λ^0 decay



In each diagram one quark of the parent emits a virtual W^- and converts to a lighter quark; spectators rearrange.

(d) Universality restoration via Cabibbo mixing

In the lepton sector g_w is universal; for quarks $g_{ud} \neq g_{us} \neq g_w$. Cabibbo: weak-eigenstate d' is a unitary rotation of mass eigenstates:

$$\begin{pmatrix} d' \\ s' \end{pmatrix} = \begin{pmatrix} \cos \theta_C & \sin \theta_C \\ -\sin \theta_C & \cos \theta_C \end{pmatrix} \begin{pmatrix} d \\ s \end{pmatrix}. \quad (23)$$

Universal vertex in weak basis:

$$g_{ud} = g_w \cos \theta_C, \quad g_{us} = g_w \sin \theta_C.$$

Sum of squares:

$$g_{ud}^2 + g_{us}^2 = g_w^2 \quad (\text{universality restored}).$$

Minimum quark count: four. A 2×2 rotation requires a partner up-type pair (u, c) , predicting charm (GIM, 1970; charm discovered 1974).

Determination of θ_C . From the data:

$$\sin \theta_C = \frac{g_{us}}{g_w} \simeq 0.221, \quad \cos \theta_C = \frac{g_{ud}}{g_w} \simeq 0.974.$$

Solve:

$$\theta_C \approx \arcsin(0.221).$$

$$\boxed{\theta_C \approx 12.8^\circ}. \quad (24)$$

Unitarity check: $0.221^2 + 0.974^2 = 0.0488 + 0.949 = 0.998 \approx 1$.

Strangeness-changing rate suppressed by $\tan^2 \theta_C \approx 0.052$.

(e) Tau branching ratios from two-generation universality

W^- couples democratically to $e^- \bar{\nu}_e, \mu^- \bar{\nu}_\mu, \bar{u} d', \bar{c} s'$. Phase space $\sim m_\tau^5$ for each. Quark channels carry colour factor 3. Cabibbo unitarity: $\bar{u} d' \rightarrow \bar{u} d (\cos^2 \theta_C) + \bar{u} s (\sin^2 \theta_C)$, summing to one. Charm channels require a D meson, $m_D \approx 1865 \text{ MeV}/c^2 > m_\tau - m_\nu = 1777 \text{ MeV}/c^2$: kinematically forbidden.

Channel weights:

$$\begin{aligned}
\tau^- &\rightarrow e^- \bar{\nu}_e \nu_\tau : & 1, \\
\tau^- &\rightarrow \mu^- \bar{\nu}_\mu \nu_\tau : & 1, \\
\tau^- &\rightarrow \nu_\tau + (\bar{u}d) : & 3 \cos^2 \theta_C \approx 2.85, \\
\tau^- &\rightarrow \nu_\tau + (\bar{u}s) : & 3 \sin^2 \theta_C \approx 0.146, \\
\tau^- &\rightarrow \nu_\tau + \bar{c} \dots : & \text{forbidden.}
\end{aligned} \tag{25}$$

Total weight:

$$W_{\text{tot}} = 1 + 1 + 3(\cos^2 \theta_C + \sin^2 \theta_C) = 5.$$

$$\boxed{\text{BR}(\tau^- \rightarrow e^- \bar{\nu}_e \nu_\tau) = \frac{1}{5} = 20\% .} \tag{26}$$

$$\boxed{\text{BR}(\tau^- \rightarrow \mu^- \bar{\nu}_\mu \nu_\tau) = \frac{1}{5} = 20\% .} \tag{27}$$

$$\boxed{\text{BR}(\tau^- \rightarrow \nu_\tau + \text{hadrons}) = \frac{3}{5} = 60\% .} \tag{28}$$

Measured: 17.8%, 17.4%, $\sim 65\%$; the deficit in the leptonic modes reflects QCD enhancement ($\sim \alpha_s/\pi$) of the hadronic width.

Consequences of mixing. Mass and weak eigenstates differ, so s, c, b, t are not weak eigenstates and must decay via W to a lighter quark with strength set by CKM. Without mixing s and c would be stable; with mixing, all heavy quarks have non-zero decay widths and ordinary matter is dominated by the lightest generation u, d .